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01

having kids

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paulgraham.com/kids.html

i remember my mother telling me once when i was about 30 that she'd really enjoyed having me and my sister. my god, i thought, this woman is a saint. she not only endured all the pain we subjected her to, but actually enjoyed it? now i realize she was simply telling the truth.

she said that one reason she liked having us was that we'd been interesting to talk to. that took me by surprise when i had kids. you don't just love them. they become your friends too. they're really interesting. and while i admit small children are disastrously fond of repetition (anything worth doing once is worth doing fifty times) it's often genuinely fun to play with them. that surprised me too. playing with a 2 year old was fun when i was 2 and definitely not fun when i was 6. why would it become fun again later? but it does.

there are of course times that are pure drudgery. or worse still, terror. having kids is one of those intense types of experience that are hard to imagine unless you've had them. but it is not, as i implicitly believed before having kids, simply your dna heading for the lifeboats.

some of my worries about having kids were right, though. they definitely make you less productive. i know having kids makes some people get their act together, but if your act was already together, you're going to have less time to do it in. in particular, you're going to have to work to a schedule. kids have schedules. i'm not sure if it's because that's how kids are, or because it's the only way to integrate their lives with adults', but once you have kids, you tend to have to work on their schedule.

you will have chunks of time to work. but you can't let work spill promiscuously through your whole life, like i used to before i had kids. you're going to have to work at the same time every day, whether inspiration is flowing or not, and there are going to be times when you have to stop, even if it is.

i've been able to adapt to working this way. work, like love, finds a way. if there are only certain times it can happen, it happens at those times. so while i don't get as much done as before i had kids, i get enough done.

i hate to say this, because being ambitious has always been a part of my identity, but having kids may make one less ambitious. it hurts to see that sentence written down. i squirm to avoid it. but if there weren't something real there, why would i squirm? the fact is, once you have kids, you're probably going to care more about them than you do about yourself. and attention is a zero-sum game. only one idea at a time can be the top idea in your mind. once you have kids, it will often be your kids, and that means it will less often be some project you're working on.

i have some hacks for sailing close to this wind. for example, when i write essays, i think about what i'd want my kids to know. that drives me to get things right. and when i was writing bel, i told my kids that once i finished it i'd take them to africa. when you say that sort of thing to a little kid, they treat it as a promise. which meant i had to finish or i'd be taking away their trip to africa. maybe if i'm really lucky such tricks could put me net ahead. but the wind is there, no question.

on the other hand, what kind of wimpy ambition do you have if it won't survive having kids? do you have so little to spare?

and while having kids may be warping my present judgement, it hasn't overwritten my memory. i remember perfectly well what life was like before. well enough to miss some things a lot, like the ability to take off for some other country at a moment's notice. that was so great. why did i never do that?

see what i did there? the fact is, most of the freedom i had before kids, i never used. i paid for it in loneliness, but i never used it.

02

lies we tell kids

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my whole world was no bigger than a few friends' houses i bicycled to and some woods i ran around in. on a log scale i was midway between crib and globe. a suburban street was just the right size. but as i grew older, suburbia started to feel suffocatingly fake.

life can be pretty good at 10 or 20, but it's often frustrating at 15. this is too big a problem to solve here, but certainly one reason life sucks at 15 is that kids are trapped in a world designed for 10 year olds.

what do parents hope to protect their children from by raising them in suburbia? a friend who moved out of manhattan said merely that her 3 year old daughter "saw too much." off the top of my head, that might include: people who are high or drunk, poverty, madness, gruesome medical conditions, sexual behavior of various degrees of oddness, and violent anger.

i think it's the anger that would worry me most if i had a 3 year old. i was 29 when i moved to new york and i was surprised even then. i wouldn't want a 3 year old to see some of the disputes i saw. it would be too frightening. a lot of the things adults conceal from smaller children, they conceal because they'd be frightening, not because they want to conceal the existence of such things. misleading the child is just a byproduct.

this seems one of the most justifiable types of lying adults do to kids. but because the lies are indirect we don't keep a very strict accounting of them. parents know they've concealed the facts about sex, and many at some point sit their kids down and explain more. but few tell their kids about the differences between the real world and the cocoon they grew up in. combine this with the confidence parents try to instill in their kids, and every year you get a new crop of 18 year olds who think they know how to run the world.

don't all 18 year olds think they know how to run the world? actually this seems to be a recent innovation, no more than about 100 years old. in preindustrial times teenage kids were junior members of the adult world and comparatively well aware of their shortcomings. they could see they weren't as strong or skillful as the village smith. in past times people lied to kids about some things more than we do now, but the lies implicit in an artificial, protected environment are a recent invention. like a lot of new inventions, the rich got this first. children of kings and great magnates were the first to grow up out of touch with the world. suburbia means half the population can live like kings in that respect.

sex (and drugs)

i'd have different worries about raising teenage kids in new york. i'd worry less about what they'd see, and more about what they'd do. i went to college with a lot of kids who grew up in manhattan, and as a rule they seemed pretty jaded. they seemed to have lost their virginity at an average of about 14 and by college had tried more drugs than i'd even heard of.

the reasons parents don't want their teenage kids having sex are complex. there are some obvious dangers: pregnancy and sexually transmitted diseases. but those aren't the only reasons parents don't want their kids having sex. the average parents of a 14 year old girl would hate the idea of her having sex even if there were zero risk of pregnancy or sexually transmitted diseases.

kids can probably sense they aren't being told the whole story. after all, pregnancy and sexually transmitted diseases are just as much a problem for adults, and they have sex.

what really bothers parents about their teenage kids having sex? their dislike of the idea is so visceral it's probably inborn. but if it's inborn it should be universal, and there are plenty of societies where parents don't mind if their teenage kids have sex—indeed, where it's normal for 14 year olds to become

othetical jaded 10 year old bothers me so much is not just that he'd be annoying, but that he'd have cut off his prospects for growth so early. to be jaded you have to think you know how the world works, and any theory a 10 year old had about that would probably be a pretty narrow one.

innocence is also open-mindedness. we want kids to be innocent so they can continue to learn. paradoxical as it sounds, there are some kinds of knowledge that get in the way of other kinds of knowledge. if you're going to learn that the world is a brutal place full of people trying to take advantage of one another, you're better off learning it last. otherwise you won't bother learning much more.

very smart adults often seem unusually innocent, and i don't think this is a coincidence. i think they've deliberately avoided learning about certain things. certainly i do. i used to think i wanted to know everything. now i know i don't.

death

after sex, death is the topic adults lie most conspicuously about to kids. sex i believe they conceal because of deep taboos. but why do we conceal death from kids? probably because small children are particularly horrified by it. they want to feel safe, and death is the ultimate threat.

one of the most spectacular lies our parents told us was about the death of our first cat. over the years, as we asked for more details, they were compelled to invent more, so the story grew quite elaborate. the cat had died at the vet's office. of what? of the anaesthesia itself. why was the cat at the vet's office? to be fixed. and why had such a routine operation killed it? it wasn't the vet's fault; the cat had a congenitally weak heart; the anaesthesia was too much for it; but there was no way anyone could have known this in advance. it was not till we were in our twenties that the truth came out: my sister, then about three, had accidentally stepped on the cat and broken its back.

they didn't feel the need to tell us the cat was now happily in cat heaven. my parents never claimed that people or animals who died had "gone to a better place," or that we'd meet them again. it didn't seem to harm us.

my grandmother told us an edited version of the death of my grandfather. she said they'd been sitting reading one day, and when she said something to him, he didn't answer. he seemed to be asleep, but when she tried to rouse him, she couldn't. "he was gone." having a heart attack sounded like falling asleep. later i learned it hadn't been so neat, and the heart attack had taken most of a day to kill him.

along with such outright lies, there must have been a lot of changing the subject when death came up. i can't remember that, of course, but i can infer it from the fact that i didn't really grasp i was going to die till i was about 19. how could i have missed something so obvious for so long? now that i've seen parents managing the subject, i can see how: questions about death are gently but firmly turned aside.

on this topic, especially, they're met half-way by kids. kids often want to be lied to. they want to believe they're living in a comfortable, safe world as much as their parents want them to believe it. [4]

identity

some parents feel a strong adherence to an ethnic or religious group and want their kids to feel it too. this usually requires two different kinds of lying: the first is to tell the child that he or she is an x, and the second is whatever specific lies xes differentiate themselves by believing. [5]

telling a child they have a particular ethnic or religious identity is one of the stickiest things you can tell them. almost anything else you tell a kid, they can change their mind about later when they start to

this happens in intellectual as well as moral questions. the more confident people are, the more willing they seem to be to answer a question "i don't know." less confident people feel they have to have an answer or they'll look bad. my parents were pretty good about admitting when they didn't know things, but i must have been told a lot of lies of this type by teachers, because i rarely heard a teacher say "i don't know" till i got to college. i remember because it was so surprising to hear someone say that in front of a class.

the first hint i had that teachers weren't omniscient came in sixth grade, after my father contradicted something i'd learned in school. when i protested that the teacher had said the opposite, my father replied that the guy had no idea what he was talking about—that he was just an elementary school teacher, after all.

just a teacher? the phrase seemed almost grammatically ill-formed. didn't teachers know everything about the subjects they taught? and if not, why were they the ones teaching us?

the sad fact is, us public school teachers don't generally understand the stuff they're teaching very well. there are some sterling exceptions, but as a rule people planning to go into teaching rank academically near the bottom of the college population. so the fact that i still thought at age 11 that teachers were infallible shows what a job the system must have done on my brain.

school

what kids get taught in school is a complex mix of lies. the most excusable are those told to simplify ideas to make them easy to learn. the problem is, a lot of propaganda gets slipped into the curriculum in the name of simplification.

public school textbooks represent a compromise between what various powerful groups want kids to be told. the lies are rarely overt. usually they consist either of omissions or of over-emphasizing certain topics at the expense of others. the view of history we got in elementary school was a crude hagiography, with at least one representative of each powerful group.

the famous scientists i remember were einstein, marie curie, and george washington carver. einstein was a big deal because his work led to the atom bomb. marie curie was involved with x-rays. but i was mystified about carver. he seemed to have done stuff with peanuts.

it's obvious now that he was on the list because he was black (and for that matter that marie curie was on it because she was a woman), but as a kid i was confused for years about him. i wonder if it wouldn't have been better just to tell us the truth: that there weren't any famous black scientists. ranking george washington carver with einstein misled us not only about science, but about the obstacles blacks faced in his time.

as subjects got softer, the lies got more frequent. by the time you got to politics and recent history, what we were taught was pretty much pure propaganda. for example, we were taught to regard political leaders as saints—especially the recently martyred kennedy and king. it was astonishing to learn later that they'd both been serial womanizers, and that kennedy was a speed freak to boot. (by the time king's plagiarism emerged, i'd lost the ability to be surprised by the misdeeds of famous people.)

i doubt you could teach kids recent history without teaching them lies, because practically everyone who has anything to say about it has some kind of spin to put on it. much recent history *consists* of spin. it would probably be better just to teach them metafacts like that.

probably the biggest lie told in schools, though, is that the way to succeed is through following "the rules." in fact most such rules are just hacks to manage large groups efficiently.

notes

1 one reason i stuck with such a brutally simple word is that the lies we tell kids are probably not quite as harmless as we think. if you look at what adults told children in the past, it's shocking how much they lied to them. like us, they did it with the best intentions. so if we think we're as open as one could reasonably be with children, we're probably fooling ourselves. odds are people in 100 years will be as shocked at some of the lies we tell as we are at some of the lies people told 100 years ago.

i can't predict which these will be, and i don't want to write an essay that will seem dumb in 100 years. so instead of using special euphemisms for lies that seem excusable according to present fashions, i'm just going to call all our lies lies.

(i have omitted one type: lies told to play games with kids' credulity. these range from "make-believe," which is not really a lie because it's told with a wink, to the frightening lies told by older siblings. there's not much to say about these: i wouldn't want the first type to go away, and wouldn't expect the second type to.)

2 calaprice, alice (ed.), *the quotable einstein*, princeton university press, 1996.

3 if you ask parents why kids shouldn't swear, the less educated ones usually reply with some question-begging answer like "it's inappropriate," while the more educated ones come up with elaborate rationalizations. in fact the less educated parents seem closer to the truth.

4 as a friend with small children pointed out, it's easy for small children to consider themselves immortal, because time seems to pass so slowly for them. to a 3 year old, a day feels like a month might to an adult. so 80 years sounds to him like 2400 years would to us.

5 i realize i'm going to get endless grief for classifying religion as a type of lie. usually people skirt that issue with some equivocation implying that lies believed for a sufficiently long time by sufficiently large numbers of people are immune to the usual standards for truth. but because i can't predict which lies future generations will consider inexcusable, i can't safely omit any type we tell. yes, it seems unlikely that religion will be out of fashion in 100 years, but no more unlikely than it would have seemed to someone in 1880 that schoolchildren in 1980 would be taught that masturbation was perfectly normal and not to feel guilty about it.

6 unfortunately the payload can consist of bad customs as well as good ones. for example, there are certain qualities that some groups in america consider "acting white." in fact most of them could as accurately be called "acting japanese." there's nothing specifically white about such customs. they're common to all cultures with long traditions of living in cities. so it is probably a losing bet for a group to consider behaving the opposite way as part of its identity.

7 in this context, "issues" basically means "things we're going to lie to them about." that's why there's a special name for these topics.

8 mayle, peter, *why are we getting a divorce?*, harmony, 1988.

9 the ironic thing is, this is also the main reason kids lie to adults. if you freak out when people tell you alarming things, they won't tell you them. teenagers don't tell their parents what happened that night they were supposed to be staying at a friend's house for the same reason parents don't tell 5 year olds the truth about the thanksgiving turkey. they'd freak if they knew.

thanks to sam altman, marc andreessen, trevor blackwell, patrick collison, jessica livingston, jackie mcdonough, robert morris, and david sloo for reading drafts of this. and since there are some controversial ideas here, i should add that none of them agreed with everything in it.

03

what you'll wish you'd known

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we need to cut the standard graduation speech down to, "what someone else with your abilities can do, you can do; and don't underestimate your abilities." but as so often happens, the closer you get to the truth, the messier your sentence gets. we've taken a nice, neat (but wrong) slogan, and churned it up like a mud puddle. it doesn't make a very good speech anymore. but worse still, it doesn't tell you what to do anymore. someone with your abilities? what are your abilities?

upwind

i think the solution is to work in the other direction. instead of working back from a goal, work forward from promising situations. this is what most successful people actually do anyway.

in the graduation-speech approach, you decide where you want to be in twenty years, and then ask: what should i do now to get there? i propose instead that you don't commit to anything in the future, but just look at the options available now, and choose those that will give you the most promising range of options afterward.

it's not so important what you work on, so long as you're not wasting your time. work on things that interest you and increase your options, and worry later about which you'll take.

suppose you're a college freshman deciding whether to major in math or economics. well, math will give you more options: you can go into almost any field from math. if you major in math it will be easy to get into grad school in economics, but if you major in economics it will be hard to get into grad school in math.

flying a glider is a good metaphor here. because a glider doesn't have an engine, you can't fly into the wind without losing a lot of altitude. if you let yourself get far downwind of good places to land, your options narrow uncomfortably. as a rule you want to stay upwind. so i propose that as a replacement for "don't give up on your dreams." stay upwind.

how do you do that, though? even if math is upwind of economics, how are you supposed to know that as a high school student?

well, you don't, and that's what you need to find out. look for smart people and hard problems. smart people tend to clump together, and if you can find such a clump, it's probably worthwhile to join it. but it's not straightforward to find these, because there is a lot of faking going on.

to a newly arrived undergraduate, all university departments look much the same. the professors all seem forbiddingly intellectual and publish papers unintelligible to outsiders. but while in some fields the papers are unintelligible because they're full of hard ideas, in others they're deliberately written in an obscure way to seem as if they're saying something important. this may seem a scandalous proposition, but it has been experimentally verified, in the famous *social text* affair. suspecting that the papers published by literary theorists were often just intellectual-sounding nonsense, a physicist deliberately wrote a paper full of intellectual-sounding nonsense, and submitted it to a literary theory journal, which published it.

the best protection is always to be working on hard problems. writing novels is hard. reading novels isn't. hard means worry: if you're not worrying that something you're making will come out badly, or that you won't be able to understand something you're studying, then it isn't hard enough. there has to be suspense.

well, this seems a grim view of the world, you may think. what i'm telling you is that you should worry? yes, but it's not as bad as it sounds. it's exhilarating to overcome worries. you don't see faces much happier than people winning gold medals. and you know why they're so happy? relief.

ide whether you get in, but admissions officers, and they are nowhere near as smart. they're the ncos of the intellectual world. they can't tell how smart you are. the mere existence of prep schools is proof of that.

few parents would pay so much for their kids to go to a school that didn't improve their admissions prospects. prep schools openly say this is one of their aims. but what that means, if you stop to think about it, is that they can hack the admissions process: that they can take the very same kid and make him seem a more appealing candidate than he would if he went to the local public school.[6]

right now most of you feel your job in life is to be a promising college applicant. but that means you're designing your life to satisfy a process so mindless that there's a whole industry devoted to subverting it. no wonder you become cynical. the malaise you feel is the same that a producer of reality tv shows or a tobacco industry executive feels. and you don't even get paid a lot.

so what do you do? what you should not do is rebel. that's what i did, and it was a mistake. i didn't realize exactly what was happening to us, but i smelled a major rat. and so i just gave up. obviously the world sucked, so why bother?

when i discovered that one of our teachers was herself using cliff's notes, it seemed par for the course. surely it meant nothing to get a good grade in such a class.

in retrospect this was stupid. it was like someone getting fouled in a soccer game and saying, hey, you fouled me, that's against the rules, and walking off the field in indignation. fouls happen. the thing to do when you get fouled is not to lose your cool. just keep playing.

by putting you in this situation, society has fouled you. yes, as you suspect, a lot of the stuff you learn in your classes is crap. and yes, as you suspect, the college admissions process is largely a charade. but like many fouls, this one was unintentional.[7] so just keep playing.

rebellion is almost as stupid as obedience. in either case you let yourself be defined by what they tell you to do. the best plan, i think, is to step onto an orthogonal vector. don't just do what they tell you, and don't just refuse to. instead treat school as a day job. as day jobs go, it's pretty sweet. you're done at 3 o'clock, and you can even work on your own stuff while you're there.

curiosity

and what's your real job supposed to be? unless you're mozart, your first task is to figure that out. what are the great things to work on? where are the imaginative people? and most importantly, what are you interested in? the word "aptitude" is misleading, because it implies something innate. the most powerful sort of aptitude is a consuming interest in some question, and such interests are often acquired tastes.

a distorted version of this idea has filtered into popular culture under the name "passion." i recently saw an ad for waiters saying they wanted people with a "passion for service." the real thing is not something one could have for waiting on tables. and passion is a bad word for it. a better name would be curiosity.

kids are curious, but the curiosity i mean has a different shape from kid curiosity. kid curiosity is broad and shallow; they ask why at random about everything. in most adults this curiosity dries up entirely. it has to: you can't get anything done if you're always asking why about everything. but in ambitious adults, instead of drying up, curiosity becomes narrow and deep. the mud flat morphs into a well.

the way to get a big idea to appear in your head is not to hunt for big ideas, but to put in a lot of time on work that interests you, and in the process keep your mind open enough that a big idea can take roost. einstein, ford, and beckenbauer all used this recipe. they all knew their work like a piano player knows the keys. so when something seemed amiss to them, they had the confidence to notice it.

put in time how and on what? just pick a project that seems interesting: to master some chunk of material, or to make something, or to answer some question. choose a project that will take less than a month, and make it something you have the means to finish. do something hard enough to stretch you, but only just, especially at first. if you're deciding between two projects, choose whichever seems most fun. if one blows up in your face, start another. repeat till, like an internal combustion engine, the process becomes self-sustaining, and each project generates the next one. (this could take years.)

it may be just as well not to do a project "for school," if that will restrict you or make it seem like work. involve your friends if you want, but not too many, and only if they're not flakes. friends offer moral support (few startups are started by one person), but secrecy also has its advantages. there's something pleasing about a secret project. and you can take more risks, because no one will know if you fail.

don't worry if a project doesn't seem to be on the path to some goal you're supposed to have. paths can bend a lot more than you think. so let the path grow out the project. the most important thing is to be excited about it, because it's by doing that you learn.

don't disregard unseemly motivations. one of the most powerful is the desire to be better than other people at something. hardy said that's what got him started, and i think the only unusual thing about him is that he admitted it. another powerful motivator is the desire to do, or know, things you're not supposed to. closely related is the desire to do something audacious. sixteen year olds aren't supposed to write novels. so if you try, anything you achieve is on the plus side of the ledger; if you fail utterly, you're doing no worse than expectations.[8]

beware of bad models. especially when they excuse laziness. when i was in high school i used to write "existentialist" short stories like ones i'd seen by famous writers. my stories didn't have a lot of plot, but they were very deep. and they were less work to write than entertaining ones would have been. i should have known that was a danger sign. and in fact i found my stories pretty boring; what excited me was the idea of writing serious, intellectual stuff like the famous writers.

now i have enough experience to realize that those famous writers actually sucked. plenty of famous people do; in the short term, the quality of one's work is only a small component of fame. i should have been less worried about doing something that seemed cool, and just done something i liked. that's the actual road to coolness anyway.

a key ingredient in many projects, almost a project on its own, is to find good books. most books are bad. nearly all textbooks are bad.[9] so don't assume a subject is to be learned from whatever book on it happens to be closest. you have to search actively for the tiny number of good books.

the important thing is to get out there and do stuff. instead of waiting to be taught, go out and learn.

your life doesn't have to be shaped by admissions officers. it could be shaped by your own curiosity. it is for all ambitious adults. and you don't have to wait to start. in fact, you don't have to wait to be an adult. there's no switch inside you that magically flips when you turn a certain age or graduate from some institution. you start being an adult when you decide to take responsibility for your life. you can do that at any age.[10]

(i'm not saying public school kids are smarter than preppies, just that they are within any given college. that follows necessarily if you agree prep schools improve kids' admissions prospects.)

7 why does society foul you? indifference, mainly. there are simply no outside forces pushing high school to be good. the air traffic control system works because planes would crash otherwise. businesses have to deliver because otherwise competitors would take their customers. but no planes crash if your school sucks, and it has no competitors. high school isn't evil; it's random; but random is pretty bad.

8 and then of course there is money. it's not a big factor in high school, because you can't do much that anyone wants. but a lot of great things were created mainly to make money. samuel johnson said "no man but a blockhead ever wrote except for money." (many hope he was exaggerating.)

9 even college textbooks are bad. when you get to college, you'll find that (with a few stellar exceptions) the textbooks are not written by the leading scholars in the field they describe. writing college textbooks is unpleasant work, done mostly by people who need the money. it's unpleasant because the publishers exert so much control, and there are few things worse than close supervision by someone who doesn't understand what you're doing. this phenomenon is apparently even worse in the production of high school textbooks.

10 your teachers are always telling you to behave like adults. i wonder if they'd like it if you did. you may be loud and disorganized, but you're very docile compared to adults. if you actually started acting like adults, it would be just as if a bunch of adults had been transposed into your bodies. imagine the reaction of an fbi agent or taxi driver or reporter to being told they had to ask permission to go the bathroom, and only one person could go at a time. to say nothing of the things you're taught. if a bunch of actual adults suddenly found themselves trapped in high school, the first thing they'd do is form a union and renegotiate all the rules with the administration.

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04

early work

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paulgraham.com/early.html

but even if silicon valley's encouraging attitude is rooted in self-interest, it has over time actually grown into a sort of benevolence. encouraging startups has been practiced for so long that it has become a custom. now it just seems that that's what one does with startups.

maybe silicon valley is too optimistic. maybe it's too easily fooled by impostors. many less optimistic journalists want to believe that. but the lists of impostors they cite are suspiciously short, and plagued with asterisks. [2] if you use revenue as the test, silicon valley's optimism seems better tuned than the rest of the world's. and because it works, it will spread.

there's a lot more to new ideas than new startup ideas, of course. the fear of making something lame holds people back in every field. but silicon valley shows how quickly customs can evolve to support new ideas. and that in turn proves that dismissing new ideas is not so deeply rooted in human nature that it can't be unlearnt.

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unfortunately, if you want to do new things, you'll face a force more powerful than other people's skepticism: your own skepticism. you too will judge your early work too harshly. how do you avoid that?

this is a difficult problem, because you don't want to completely eliminate your horror of making something lame. that's what steers you toward doing good work. you just want to turn it off temporarily, the way a painkiller temporarily turns off pain.

people have already discovered several techniques that work. hardy mentions two in *a mathematician's apology*:

> good work is not done by "humble" men. it is one of the first duties of a professor, for example, in any subject, to exaggerate a little both the importance of his subject and his importance in it.

if you overestimate the importance of what you're working on, that will compensate for your mistakenly harsh judgment of your initial results. if you look at something that's 20% of the way to a goal worth 100 and conclude that it's 10% of the way to a goal worth 200, your estimate of its expected value is correct even though both components are wrong.

it also helps, as hardy suggests, to be slightly overconfident. i've noticed in many fields that the most successful people are slightly overconfident. on the face of it this seems implausible. surely it would be optimal to have exactly the right estimate of one's abilities. how could it be an advantage to be mistaken? because this error compensates for other sources of error in the opposite direction: being slightly overconfident armors you against both other people's skepticism and your own.

ignorance has a similar effect. it's safe to make the mistake of judging early work as finished work if you're a sufficiently lax judge of finished work. i doubt it's possible to cultivate this kind of ignorance, but empirically it's a real advantage, especially for the young.

another way to get through the lame phase of ambitious projects is to surround yourself with the right people — to create an eddy in the social headwind. but it's not enough to collect people who are always encouraging. you'd learn to discount that. you need colleagues who can actually tell an ugly duckling from a baby swan. the people best able to do this are those working on similar projects of their own, which is why university departments and research labs work so well. you don't need institutions to collect colleagues. they naturally coalesce, given the chance. but it's very much worth accelerating this process by seeking out other people trying to do new things.

ion 1. we're already evolving better customs, and we can already see signs of how big the payoff will be.

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notes

1 this assumption may be too conservative. there is some evidence that historically the bay area has attracted a different sort of person than, say, new york city.

2 one of their great favorites is theranos. but the most conspicuous feature of theranos's cap table is the absence of silicon valley firms. journalists were fooled by theranos, but silicon valley investors weren't.

3 i made two mistakes about teachers when i was younger. i cared more about professors' research than their reputations as teachers, and i was also wrong about what it meant to be a good teacher. i thought it simply meant to be good at explaining things.

4 patrick collison points out that you can go past treating something as a hack in the sense of a prototype and onward to the sense of the word that means something closer to a practical joke:

> i think there may be something related to being a hack that can be powerful — the idea of making the tenuousness and implausibility *a feature*. "yes, it's a bit ridiculous, right? i'm just trying to see how far such a naive approach can get." yc seemed to me to have this characteristic.

5 much of the advantage of switching from physical to digital media is not the software per se but that it lets you start something new with little upfront commitment.

6 john carmack adds:

> the value of a medium without a vast gulf between the early work and the final work is exemplified in game mods. the original quake game was a golden age for mods, because everything was very flexible, but so crude due to technical limitations, that quick hacks to try out a gameplay idea weren't all *that* far from the official game. many careers were born from that, but as the commercial game quality improved over the years, it became almost a full time job to make a successful mod that would be appreciated by the community. this was dramatically reversed with minecraft and later roblox, where the entire esthetic of the experience was so explicitly crude that innovative gameplay concepts became the overriding value. these "crude" game mods by single authors are now often bigger deals than massive professional teams' work.

7 lisa randall suggests that we

> treat new things as experiments. that way there's no such thing as failing, since you learn something no matter what. you treat it like an experiment in the sense that if it really rules something out, you give up and move on, but if there's some way to vary it to make it work better, go ahead and do that

8 michael nielsen points out that the internet has made this easier, because you can see programmers' first commits, musicians' first videos, and so on.

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05

being a noob

january 2020
paulgraham.com/noob.html

06

the need to read

november 2022
paulgraham.com/read.html

07

beyond smart

october 2021
paulgraham.com/smart.html

y about one of the most important: an obsessive interest in a particular topic. and this can definitely be cultivated.

another quality you need in order to discover new ideas is independent-mindedness. i wouldn't want to claim that this is distinct from intelligence — i'd be reluctant to call someone smart who wasn't independent-minded — but though largely inborn, this quality seems to be something that can be cultivated to some extent.

there are general techniques for having new ideas — for example, for working on your own projects and for overcoming the obstacles you face with early work — and these can all be learned. some of them can be learned by societies. and there are also collections of techniques for generating specific types of new ideas, like startup ideas and essay topics.

and of course there are a lot of fairly mundane ingredients in discovering new ideas, like working hard, getting enough sleep, avoiding certain kinds of stress, having the right colleagues, and finding tricks for working on what you want even when it's not what you're supposed to be working on. anything that prevents people from doing great work has an inverse that helps them to. and this class of ingredients is not as boring as it might seem at first. for example, having new ideas is generally associated with youth. but perhaps it's not youth per se that yields new ideas, but specific things that come with youth, like good health and lack of responsibilities. investigating this might lead to strategies that will help people of any age to have better ideas.

one of the most surprising ingredients in having new ideas is writing ability. there's a class of new ideas that are best discovered by writing essays and books. and that "by" is deliberate: you don't think of the ideas first, and then merely write them down. there is a kind of thinking that one does by writing, and if you're clumsy at writing, or don't enjoy doing it, that will get in your way if you try to do this kind of thinking. [4]

i predict the gap between intelligence and new ideas will turn out to be an interesting place. if we think of this gap merely as a measure of unrealized potential, it becomes a sort of wasteland that we try to hurry through with our eyes averted. but if we flip the question, and start inquiring into the other ingredients in new ideas that it implies must exist, we can mine this gap for discoveries about discovery.

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notes

1 what wins in conversation depends on who with. it ranges from mere aggressiveness at the bottom, through quick-wittedness in the middle, to something closer to actual intelligence at the top, though probably always with some component of quick-wittedness.

2 just as intelligence isn't the only ingredient in having new ideas, having new ideas isn't the only thing intelligence is useful for. it's also useful, for example, in diagnosing problems and figuring out how to fix them. both overlap with having new ideas, but both have an end that doesn't.

those ways of using intelligence are much more common than having new ideas. and in such cases intelligence is even harder to distinguish from its consequences.

3 some would attribute the difference between intelligence and having new ideas to "creativity," but this doesn't seem a very useful term. as well as being pretty vague, it's shifted half a frame sideways from what we care

08

copy what you like

july 2006
paulgraham.com/copy.html

they're easier to see, and of course easier to copy too. for example, most painters in the eighteenth and nineteenth centuries used brownish colors. they were imitating the great painters of the renaissance, whose paintings by that time were brown with dirt. those paintings have since been cleaned, revealing brilliant colors; their imitators are of course still brown.

it was painting, incidentally, that cured me of copying the wrong things. halfway through grad school i decided i wanted to try being a painter, and the art world was so manifestly corrupt that it snapped the leash of credulity. these people made philosophy professors seem as scrupulous as mathematicians. it was so clearly a choice of doing good work xor being an insider that i was forced to see the distinction. it's there to some degree in almost every field, but i had till then managed to avoid facing it.

that was one of the most valuable things i learned from painting: you have to figure out for yourself what's good. you can't trust authorities. they'll lie to you on this one.

comment on this essay.

09

the island test

july 2006
paulgraham.com/island.html

there is a point where i'll do without books. i was walking in some steep mountains once, and decided i'd rather just think, if i was bored, rather than carry a single unnecessary ounce. it wasn't so bad. i found i could entertain myself by having ideas instead of reading other people's. if you stop eating jam, fruit starts to taste better.

so maybe i'll try not bringing books on some future trip. they're going to have to pry the plugs out of my cold, dead ears, however.

10

the top idea in your mind

paulgraham.com/top.html

ing money is almost by definition an attention sink. the other is disputes. these too are engaging in the wrong way: they have the same velcro-like shape as genuinely interesting ideas, but without the substance. so avoid disputes if you want to get real work done. [3]

even newton fell into this trap. after publishing his theory of colors in 1672 he found himself distracted by disputes for years, finally concluding that the only solution was to stop publishing:

> i see i have made myself a slave to philosophy, but if i get free of mr linus's business i will resolutely bid adew to it eternally, excepting what i do for my privat satisfaction or leave to come out after me. for i see a man must either resolve to put out nothing new or become a slave to defend it. [4]

linus and his students at liege were among the more tenacious critics. newton's biographer westfall seems to feel he was overreacting:

> recall that at the time he wrote, newton's "slavery" consisted of five replies to liege, totalling fourteen printed pages, over the course of a year.

i'm more sympathetic to newton. the problem was not the 14 pages, but the pain of having this stupid controversy constantly reintroduced as the top idea in a mind that wanted so eagerly to think about other things.

turning the other cheek turns out to have selfish advantages. someone who does you an injury hurts you twice: first by the injury itself, and second by taking up your time afterward thinking about it. if you learn to ignore injuries you can at least avoid the second half. i've found i can to some extent avoid thinking about nasty things people have done to me by telling myself: this doesn't deserve space in my head. i'm always delighted to find i've forgotten the details of disputes, because that means i hadn't been thinking about them. my wife thinks i'm more forgiving than she is, but my motives are purely selfish.

i suspect a lot of people aren't sure what's the top idea in their mind at any given time. i'm often mistaken about it. i tend to think it's the idea i'd want to be the top one, rather than the one that is. but it's easy to figure this out: just take a shower. what topic do your thoughts keep returning to? if it's not what you want to be thinking about, you may want to change something.

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notes

1 no doubt there are already names for this type of thinking, but i call it "ambient thought."

2 this was made particularly clear in our case, because neither of the funds we raised was difficult, and yet in both cases the process dragged on for months. moving large amounts of money around is never something people treat casually. the attention required increases with the amount—maybe not linearly, but definitely monotonically.

3 corollary: avoid becoming an administrator, or your job will consist of dealing with money and disputes.

4 letter to oldenburg, quoted in westfall, richard, *life of isaac newton*, p. 107.

thanks to sam altman, patrick collison, jessica livingston, and robert morris for reading drafts of this.

11

good and bad procrastination

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paulgraham.com/procrastination.html

i've wondered a lot about why startups are most productive at the very beginning, when they're just a couple guys in an apartment. the main reason may be that there's no one to interrupt them yet. in theory it's good when the founders finally get enough money to hire people to do some of the work for them. but it may be better to be overworked than interrupted. once you dilute a startup with ordinary office workers — with type-b procrastinators — the whole company starts to resonate at their frequency. they're interrupt-driven, and soon you are too.

errands are so effective at killing great projects that a lot of people use them for that purpose. someone who has decided to write a novel, for example, will suddenly find that the house needs cleaning. people who fail to write novels don't do it by sitting in front of a blank page for days without writing anything. they do it by feeding the cat, going out to buy something they need for their apartment, meeting a friend for coffee, checking email. "i don't have time to work," they say. and they don't; they've made sure of that.

(there's also a variant where one has no place to work. the cure is to visit the places where famous people worked, and see how unsuitable they were.)

i've used both these excuses at one time or another. i've learned a lot of tricks for making myself work over the last 20 years, but even now i don't win consistently. some days i get real work done. other days are eaten up by errands. and i know it's usually my fault: i *let* errands eat up the day, to avoid facing some hard problem.

the most dangerous form of procrastination is unacknowledged type-b procrastination, because it doesn't feel like procrastination. you're "getting things done." just the wrong things.

any advice about procrastination that concentrates on crossing things off your to-do list is not only incomplete, but positively misleading, if it doesn't consider the possibility that the to-do list is itself a form of type-b procrastination. in fact, possibility is too weak a word. nearly everyone's is. unless you're working on the biggest things you could be working on, you're type-b procrastinating, no matter how much you're getting done.

in his famous essay you and your research (which i recommend to anyone ambitious, no matter what they're working on), richard hamming suggests that you ask yourself three questions:

1. what are the most important problems in your field?
2. are you working on one of them?
3. why not?

hamming was at bell labs when he started asking such questions. in principle anyone there ought to have been able to work on the most important problems in their field. perhaps not everyone can make an equally dramatic mark on the world; i don't know; but whatever your capacities, there are projects that stretch them. so hamming's exercise can be generalized to:

> what's the best thing you could be working on, and why aren't you?

most people will shy away from this question. i shy away from it myself; i see it there on the page and quickly move on to the next sentence. hamming used to go around actually asking people this, and it didn't make him popular. but it's a question anyone ambitious should face.

the trouble is, you may end up hooking a very big fish with this bait. to do good work, you need to do more than find good projects. once you've found them, you have to get yourself to work on them, and that can be hard. the bigger the problem, the harder it is to get yourself to work on it.

12

how to disagree

march 2008
paulgraham.com/disagree.html

> i can't believe the author dismisses intelligent design in such a cavalier fashion.

though better than attacking the author, this is still a weak form of disagreement. it matters much more whether the author is wrong or right than what his tone is. especially since tone is so hard to judge. someone who has a chip on their shoulder about some topic might be offended by a tone that to other readers seemed neutral.

so if the worst thing you can say about something is to criticize its tone, you're not saying much. is the author flippant, but correct? better that than grave and wrong. and if the author is incorrect somewhere, say where.

dh3. contradiction.

in this stage we finally get responses to what was said, rather than how or by whom. the lowest form of response to an argument is simply to state the opposing case, with little or no supporting evidence.

this is often combined with dh2 statements, as in:

> i can't believe the author dismisses intelligent design in such a cavalier fashion. intelligent design is a legitimate scientific theory.

contradiction can sometimes have some weight. sometimes merely seeing the opposing case stated explicitly is enough to see that it's right. but usually evidence will help.

dh4. counterargument.

at level 4 we reach the first form of convincing disagreement: counterargument. forms up to this point can usually be ignored as proving nothing. counterargument might prove something. the problem is, it's hard to say exactly what.

counterargument is contradiction plus reasoning and/or evidence. when aimed squarely at the original argument, it can be convincing. but unfortunately it's common for counterarguments to be aimed at something slightly different. more often than not, two people arguing passionately about something are actually arguing about two different things. sometimes they even agree with one another, but are so caught up in their squabble they don't realize it.

there could be a legitimate reason for arguing against something slightly different from what the original author said: when you feel they missed the heart of the matter. but when you do that, you should say explicitly you're doing it.

dh5. refutation.

the most convincing form of disagreement is refutation. it's also the rarest, because it's the most work. indeed, the disagreement hierarchy forms a kind of pyramid, in the sense that the higher you go the fewer instances you find.

to refute someone you probably have to quote them. you have to find a "smoking gun," a passage in whatever you disagree with that you feel is mistaken, and then explain why it's mistaken. if you can't find an actual quote to disagree with, you may be arguing with a straw man.

while refutation generally entails quoting, quoting doesn't necessarily imply refutation. some writers quote parts of things they disagree with to give the appearance of legitimate refutation, then follow with a response as low as dh3 or even dh0.

dh6. refuting the central point.

13

keep your identity small

february 2009
paulgraham.com/identity.html

ome part of their identity, then all other things being equal, the best plan is to let as few things into your identity as possible. [2]

most people reading this will already be fairly tolerant. but there is a step beyond thinking of yourself as x but tolerating y: not even to consider yourself an x. the more labels you have for yourself, the dumber they make you.

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notes

1 when that happens, it tends to happen fast, like a core going critical. the threshold for participating goes down to zero, which brings in more people. and they tend to say incendiary things, which draw more and angrier counterarguments.

2 there may be some things it's a net win to include in your identity. for example, being a scientist. but arguably that is more of a placeholder than an actual label—like putting nmi on a form that asks for your middle initial—because it doesn't commit you to believing anything in particular. a scientist isn't committed to believing in natural selection in the same way a biblical literalist is committed to rejecting it. all he's committed to is following the evidence wherever it leads.

considering yourself a scientist is equivalent to putting a sign in a cupboard saying "this cupboard must be kept empty." yes, strictly speaking, you're putting something in the cupboard, but not in the ordinary sense.

thanks to sam altman, trevor blackwell, paul buchheit, and robert morris for reading drafts of this.

14

maker's schedule, manager's schedule

paulgraham.com/makersschedule.html

chunk of time to meet founders we've funded. these chunks of time are at the end of my working day, and i wrote a signup program that ensures all the appointments within a given set of office hours are clustered at the end. because they come at the end of my day these meetings are never an interruption. (unless their working day ends at the same time as mine, the meeting presumably interrupts theirs, but since they made the appointment it must be worth it to them.) during busy periods, office hours sometimes get long enough that they compress the day, but they never interrupt it.

when we were working on our own startup, back in the 90s, i evolved another trick for partitioning the day. i used to program from dinner till about 3 am every day, because at night no one could interrupt me. then i'd sleep till about 11 am, and come in and work until dinner on what i called "business stuff." i never thought of it in these terms, but in effect i had two workdays each day, one on the manager's schedule and one on the maker's.

when you're operating on the manager's schedule you can do something you'd never want to do on the maker's: you can have speculative meetings. you can meet someone just to get to know one another. if you have an empty slot in your schedule, why not? maybe it will turn out you can help one another in some way.

business people in silicon valley (and the whole world, for that matter) have speculative meetings all the time. they're effectively free if you're on the manager's schedule. they're so common that there's distinctive language for proposing them: saying that you want to "grab coffee," for example.

speculative meetings are terribly costly if you're on the maker's schedule, though. which puts us in something of a bind. everyone assumes that, like other investors, we run on the manager's schedule. so they introduce us to someone they think we ought to meet, or send us an email proposing we grab coffee. at this point we have two options, neither of them good: we can meet with them, and lose half a day's work; or we can try to avoid meeting them, and probably offend them.

till recently we weren't clear in our own minds about the source of the problem. we just took it for granted that we had to either blow our schedules or offend people. but now that i've realized what's going on, perhaps there's a third option: to write something explaining the two types of schedule. maybe eventually, if the conflict between the manager's schedule and the maker's schedule starts to be more widely understood, it will become less of a problem.

those of us on the maker's schedule are willing to compromise. we know we have to have some number of meetings. all we ask from those on the manager's schedule is that they understand the cost.

thanks to sam altman, trevor blackwell, paul buchheit, jessica livingston, and robert morris for reading drafts of this.

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